

# Jean-Baptiste Himbert

Doctoral Researcher in Applied Mathematics

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## Research Experience

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- École Nationale des Ponts et Chaussées / Inria MATHERIALS / IPParis** Sep. 2025 – Present  
*PhD in Applied Mathematics — Advisors: Tony Lelièvre and Gabriel Stoltz*
- PhD topic: Machine Learning for Enhanced Sampling in Molecular Dynamics
  - Develop spectral methods for collective variable discovery; manuscript in preparation: *Reparametrization of Slow Modes for Collective Variable Discovery in Molecular Dynamics*.
  - Study convergence of adaptive enhanced-sampling and biasing methods in molecular dynamics.
  - Develop ML-augmented combinatorial-optimization methods for finding ground-state configurations in alloys.
  - Teach graduate-level classes in Statistics and Analysis.
- École Normale Supérieure (ENS Ulm) — Center of Data Science** Apr. 2024 – Sept. 2024  
*Research Intern — Supervisor: Stéphane Mallat*
- Developed multiscale generative models using wavelet/scattering transforms.
  - Applied them to reproduce the multiscale statistics of Lagrangian turbulence trajectories.
- Flatiron Institute — Center for Computational Mathematics** Jun. 2024 - Aug. 2024  
*Research Intern — Supervisors: François Lanusse, Alberto Bietti and Stéphane Mallat*
- Developed multiscale ViT tokenizers for generative modelling of astrophysical data.
  - Contributed to Polymathic AI's work on foundation models for scientific simulation data.
- Inria Chile — OceanAI Project** Apr. 2023 – Aug. 2023  
*Research Intern — Supervisor: Luis Marti*
- Developed hierarchical ViT/CNN models for fine-grained plankton-image classification.
  - Built tooling for large-scale, reproducible HPC/GPU experiments on the Grid'5000 infrastructure.

## Professional Experience

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- SNCF — DTIPG / Tech4Rail** Oct. 2024 – Jul. 2025  
*Consultant — 9-month strategic mission during Collège des Ingénieurs MBA*
- Conducted a techno-economic study of train-localization technologies within the European Tech4Rail programme.
  - Studied perception-based alternatives to conventional train localization for deployment in SIL4 safety-critical systems.
- BearingPoint** Jun. – Aug. 2022  
*Consultant Intern — Government & Public Sector*
- Designed and facilitated strategic and project-management workshops for the French Ministry of the Interior.
- French Army — RSMA Mayotte** Jan. – Mar. 2021  
*Platoon Leader*
- Led training and field exercises for a platoon of 20-30 recruits.

## Education

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- Collège des Ingénieurs** — Master of Business Administration (MBA) 2024 – 2025
- École des Ponts / ENS Paris-Saclay** — MVA — Mathematics, Vision, Learning — Highest Honors 2023 – 2024
- Research-oriented MSc in machine learning and applied mathematics; coursework in statistics, deep learning, optimization, geometry and reinforcement learning.
- École Polytechnique** — Diplôme d'Ingénieur, Applied Mathematics — GPA 3.91/4 2020 – 2023
- Entrance rank: 9th/~7,000 nationally; Graduation rank: 26th/500.
  - Completed advanced coursework in applied mathematics, machine learning, optimization, physics and computational chemistry.
- Université Paris Nanterre** — Bachelor's degree in Philosophy (alongside École Polytechnique) 2021 – 2022

## Selected Competitions

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- ADIA Lab / CrunchDAO — Structural Break Challenge — Top 5% of participants** 2026  
Developed large-scale statistical and ML methods for structural-break detection in univariate time series.
- Conseil d'État AI Hackathon — 1st place (€5,000)** 2024  
Built a multimodal AI pipeline to improve accessibility of judicial decisions.

## Skills

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**ML / Scientific computing:** Python, PyTorch, NumPy/SciPy, Julia, HPC/GPU, OpenMM, NAMD, Plumed, Slurm, Linux, Git, Docker, Weights & Biases, LaTeX.

**Languages:** French (native), English (fluent), Spanish (intermediate). **Interests:** Triathlon, Ski, Karate, Chess.